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Module Title : Machine Learning and Artificial Intelligence

Duration : 5 days

Outline

1. **Understanding the Big Picture**
   - Artificial Intelligence (AI) Overview
   - What is Machine Learning (ML)?
   - AI vs ML vs Data Science
   - Relationship between Deep Learning (DL) and Machine Learning
   - Practical Use Cases
   - Concepts and Terms
   - Tools/Platforms for ML, DL, and AI
   - Machine Learning Project End-to-End Pipeline
   - Scalable ML/AI: Big Data and Cloud in the Ecosystem

2. **Environment for Experiments**
   - Installing Anaconda
   - Setting up Jupyter Notebook & Google Colab
   - Experiencing Notebooks
   - Key Python Syntax Recap

3. **Interactive Data Exploration using Google Colab**
   - Using Google Colab for ML Experiments
   - Exploring Cloud Storage Options for ML Datasets
   - Running ML Models on Google Colab
   - Making API Calls from Notebooks

4. **Key Statistics**

- Summary Statistics
- Exploratory Data Analysis
- Numerical Computation using Python

5. Data Visualization

- Overview
- Using Matplotlib
- Working with Seaborn
- Key Types of Plots
- Exploratory Analysis using Seaborn

6. Acquiring & Preparing Data

- Content Acquisition Overview
- Working with Beautiful Soup
- Acquiring Data using REST-based APIs
- Data Cleaning & Wrangling using Pandas
- Handling Missing Values and Outliers
- Cleansing Twitter Data
- Performing Twitter Sentiment Analysis

7. Feature Engineering

- What is Feature Engineering?
- Why Feature Engineering?
- How to Apply Feature Engineering?
- Discussions on Various Scenarios

8. Machine Learning using Scikit-Learn

- Types of Machine Learning
- Key Algorithms in Machine Learning
- Practical Applications of Machine Learning
- Various Frameworks/Libraries Popular for ML
- Concepts and Terms

- Why Scikit-Learn?

9. Supervised Machine Learning

- Key Classification Algorithms
- Naïve Bayes Classifier
- Confusion Matrix
- Accuracy
- Key Regression Algorithms
- Linear, Logistic Regressions
- Gradient Descent
- Loss Function
- Bias vs Variance Tradeoff
- Evaluating Models

10. Unsupervised Machine Learning

- Why Unsupervised Learning is Important?
- Where It Can Be Applied?
- Principal Component Analysis
- Performing Clustering of Data

11. Introduction to Deep Learning

- What is Deep Learning?
- Relationship between Deep Learning and Machine Learning
- Deep Learning Use Cases
- Concepts and Terms
- How to Implement Deep Learning?
- Various Libraries, Pros & Cons

12. Artificial Neural Networks

- Introduction to Neural Networks
- Introduction to Perceptron
- Neural Network Activation Functions
- Basic Neural Nets
- Concepts

13. TensorFlow API

- What is TensorFlow?
- Installing TensorFlow
- TensorFlow Graphs
- Variables and Placeholders
- Activation Functions

14. Convolutional Neural Networks (CNN)

- CNN History
- Understanding CNNs
- CNN Applications

15. Conversational AI

- Introduction to Chatbots
- Key Options Available
- Building a Chatbot
- Hands-on Exercise using DialogFlow API